

# Scorecard Update

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## Overview

This vignette shows the data logging process of each round and shows trends and metric averages

## Set-up Environment

### Attach Packages

```
library(golf)
library(tidyverse)
library(lme4)
library(mgcv)
library(brms)
library(DBI)
library(RSQLite)
# library(blastula)
# library(keyring)
# library(mailR)
library(emayili)
```

## Record New Scorecard

### Input the Scores Data

```
round_course <- 'Randolph North'
round_date <- '2026-04-05'
round_teens <- 'blue'

hole_scores <- c(4, 6, 5, 5, 5, 4, 6, 4, 6,
                 5, 3, 5, 5, 4, 6, 6, 4, 5)

FIRs <- c(0, 0, 1, 1, 0, 0, 1, 0, 0,
          1, 0, 1, 0, 1, 0, 0, 0, 0)

GIRs <- c(0, 0, 1, rep(0, 6),
          0, 1, 0, 0, 1, rep(0, 4))
```

```

putts_rec <- c(1, 2, 2, 1, 2, 2, 2, 2, 1,
              2, 2, 2, 2, 2, 2, 1, 2, 1, 2)

chips_rec <- c(1, 2, 1, 2, 1, 1, 2, 1, 3,
              1, 0, 1, 1, 0, 1, 1, 1, 1)

penalties_rec <- c(rep(0, 14), 2, 0, 0, 0)

tee_clubs <- c('4', 'D', 'D', 'D', 'D', '9', 'D', '5', 'D',
              'D', '6', 'D', 'D', 'D', '6', 'D', 'D', 'D')

index <- 11.0

```

## Specify Course and Tees

```

if ( length(hole_scores) > 0 ) {
  Card <- golf::get_course(course = round_course, date = round_date, tees = round_tees)
}

```

## Get Scoring Metrics

```

if ( length(hole_scores) > 0 ) {
  Card <- golf::log_score(Scorecard = Card,
                        hole_by_hole = hole_scores,
                        GHIN = 10526424,
                        index = index,
                        FIR = FIRs,
                        GIR = GIRs,
                        putts = putts_rec,
                        chips = chips_rec,
                        penalties = penalties_rec,
                        tee_club = tee_clubs)
}

```

## Update the db

### Update the Players Table

```

con <- golf::get_db_connection()

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC")) != NULL ) {
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
}

```

```

    ) {

DBI::dbAppendTable(conn = con,
  name = 'players',
  value = players <- Card |>
    dplyr::select(GHIN, index, date) |>
    dplyr::distinct() |>
    dplyr::rename(handicap_index = index) |>
    dplyr::mutate(date = as.character(date))
  )
}

```

## Update the Courses Table

```

con <- golf::get_db_connection()

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC"))
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
DBI::dbAppendTable(conn = con,
  name = 'courses',
  value = course <- Card |>
    dplyr::select(course, tees, to_par, slope, course_rating, hole, yds, par, hole_handicap) |>
    dplyr::distinct() |>
    dplyr::rename(course_name = course) |>
    dplyr::mutate(date = as.character(date))
  )
}

```

## Update the Rounds Table

```

con <- golf::get_db_connection()

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC"))
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
DBI::dbAppendTable(conn = con,
  name = 'rounds',
  value = Card |>

```

```

    dplyr::select(-to_par, -slope, -course_rating, -yds, -par) |>
    dplyr::distinct() |>
    dplyr::rename(handicap_index = index) |>
    dplyr::rename(course_name = course) |>
    dplyr::mutate(date = as.character(date))
  )
}

```

## Get Club Metrics df

```
club_metrics_df <- golf::get_tracked_shots_data_shape(round_date = round_date)
```

## Annotate Club Metrics

```

club_choice <- c(
  '4', 'PW', 'PW',
  'D', '4', 'SW', 'LW',
  'D', '3W', 'SW',
  'D', '7', 'PW', 'P',
  'D', '8', 'PW',
  '9', 'GW',
  'D', 'PW', 'PW', 'P',
  '5', 'SW',
  'D', '6', 'SW', 'GW', 'P',

  'D', 'GW', 'SW',
  '6',
  'D', 'GW', 'SW',
  'D', '4', 'SW',
  'D', 'PW',
  '6', '6', 'GW',
  'D', '3W', 'SW', 'GW',
  'D', '4', 'P',
  'D', '7', 'SW'
)

dist_to_target <- c(
  220, 140, 10,
  270, 115, 30, 15,
  270, 250, 30,
  270, 175, 20, 10,
  270, 165, 13,
  145, 19,
  270, 135, 25, 15,
  205, 40,
  270, 190, 65, 21, 11,

  270, 125, 24,
  190,

```

```

270, 120, 45,
270, 120, 90,
270, 140,
190, 190, 10,
270, 250, 100, 10,
270, 150, 5,
270, 154, 32
)

yds <- c(
  219, 131, 8,
  308, 82, 16, 16,
  272, 260, 28,
  259, 168, 20, 10,
  255, 167, 17,
  123, 8,
  257, 97, 39, 19,
  210, 36,
  279, 165, 33, 21, 10,

  290, 102, 23,
  188,
  267, 69, 49,
  304, 159, 90,
  286, 139,
  180, 184, 5,
  248, 222, 114, 6,
  273, 144, 4,
  265, 142, 16
)

lie_type <- c(
  'tee', 'rough', 'fairway',
  'tee', 'rough', 'sand', 'sand',
  'tee', 'fairway', 'rough',
  'tee', 'fairway', 'fairway', 'fairway',
  'tee', 'rough', 'fairway',
  'tee', 'rough',
  'tee', 'fairway', 'fairway', 'fairway',
  'tee', 'rough',
  'tee', 'fairway', 'rough', 'rough', 'fairway',

  'tee', 'rough', 'fairway',
  'tee',
  'tee', 'fairway', 'fairway',
  'tee', 'rough', 'rough',
  'tee', 'fairway',
  'tee', 'tee', 'fairway',
  'tee', 'rough', 'rough', 'fairway',
  'tee', 'rough', 'fairway',
  'tee', 'rough', 'rough'
)

```

```

target_status <- c(
  'no', 'no', 'yes',
  'no', 'no', 'no', 'yes',
  'yes', 'no', 'yes',
  'yes', 'no', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'yes',
  'yes', 'no', 'no', 'yes',
  'no', 'yes',
  'yes', 'no', 'no', 'no', 'yes',

  'no', 'no', 'yes',
  'yes',
  'yes', 'no', 'yes',
  'no', 'no', 'yes',
  'yes', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'no', 'yes'
)

location <- c(
  'right', 'right', 'on_target',
  'left', 'short', 'short', 'on_target',
  'on_target', 'right', 'on_target',
  'on_target', 'long', 'long', 'on_target',
  'right', 'short', 'on_target',
  'short', 'on_target',
  'on_target', 'short', 'long', 'on_target',
  'long', 'on_target',
  'on_target', 'right', 'short', 'short', 'on_target',

  'right', 'short', 'on_target',
  'on_target',
  'on_target', 'short', 'on_target',
  'right', 'right', 'on_target',
  'on_target', 'on_target',
  'short', 'short', 'on_target',
  'right', 'right', 'long', 'on_target',
  'long', 'right', 'on_target',
  'right', 'short', 'on_target'
)

type_of_shot <- c(
  'tee', 'full', 'chip',
  'tee', 'punch', 'gsbunker', 'gsbunker',
  'tee', 'full', 'chip',
  'tee', 'full', 'chip', 'chip',
  'tee', 'full', 'chip',
  'tee', 'chip',
  'tee', 'full', 'chip', 'chip',
  'tee', 'chip',

```

```

'tee', 'full', 'chip', 'chip', 'chip',
'tee', 'full', 'chip',
'tee',
'tee', 'full', 'chip',
'tee', 'punch', 'full',
'tee', 'full',
'tee', 'tee', 'chip',
'tee', 'full', 'full', 'chip',
'tee', 'punch', 'chip',
'tee', 'full', 'chip'
)

```

## Get Shot Metrics

```

##      tracked_shots sum(tracked_shots)
## 1              3              55
## 2              4              55
## 3              3              55
## 4              5              55
## 5              2              55
## 6              3              55
## 7              3              55
## 8              3              55
## 9              2              55
## 10             3              55
## 11             2              55
## 12             4              55
## 13             3              55
## 14             4              55
## 15             1              55
## 16             4              55
## 17             3              55
## 18             3              55

```

## Update the Club Metrics Table

```

con <- golf::get_db_connection()

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY date")) %>%
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
  DBI::dbAppendTable(conn = con,
    name = 'club_metrics',
    value = club_metrics |>
      dplyr::mutate(date = as.character(date))
  )
}

```

```

    )
}

rm(list = ls()[which(grepl(ls(), pattern= 'con|round_course|round_date|index')==F)])

```

## Summarize Metrics

### Gather and Format

Gather and format from the database

```

con <- golf::get_db_connection()

scores <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT sub.* FROM
  (SELECT DISTINCT r.*, c.par, c.course_rating FROM rounds r
  LEFT JOIN courses c
  ON c.tees = r.tees
  AND c.course_name = r.course_name
  AND c.hole = r.hole
  AND c.hole_handicap = r.hole_handicap) AS sub
  INNER JOIN players p
  ON sub.GHIN = p.GHIN
  AND sub.handicap_index = p.handicap_index
  AND sub.date = p.date;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date),
               hole = stringr::str_extract(hole, pattern = '[0-9]{1,}') |>
               as.numeric()) |>
  dplyr::relocate(par, .after = hole) |>
  dplyr::relocate(course_rating, .after = tees) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole) |>
  dplyr::ungroup()

```

```

con <- golf::get_db_connection()

stroke_quality <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT * FROM club_metrics;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date),
               hole = stringr::str_extract(hole, pattern = '[0-9]{1,}') |>
               as.numeric()) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole, stroke) |>
  dplyr::ungroup()

```

### Compute Advanced Metrics

Compute more nuanced metrics



```

scores_sum <- scores |>
  dplyr::mutate(date_course = paste0(date,
                                     '\n',
                                     course_name, '\n',
                                     handicap_index),
               chips = dplyr::case_when(
                 grepl(date, pattern = '07-13') ~ NA_real_, TRUE ~ chips),
               `chips+putts` = chips+putts,
               FIR_opps = dplyr::case_when(par > 3 ~ 1, TRUE ~ NA),
               `Iron FIRs` = dplyr::case_when(par > 3 &
                                             grepl(tee_club, pattern = '4|5') &
                                             FIR == 1 ~ 1,
                                             TRUE ~ 0),
               `Iron FIR opps` = dplyr::case_when(par > 3 &
                                             grepl(tee_club, pattern = '4|5') ~ 1,
                                             TRUE ~ 0),
               `Driver FIRs` = dplyr::case_when(par > 3 &
                                             tee_club == 'D' &
                                             FIR == 1 ~ 1,
                                             TRUE ~ 0),
               `Driver FIR opps` = dplyr::case_when(par > 3 &
                                             tee_club == 'D' ~ 1,
                                             TRUE ~ 0),
               `Par 3 GIRs` = dplyr::case_when(par == 3 &
                                             GIR == 1 ~ 1,
                                             TRUE ~ 0),
               greenie_putts = dplyr::case_when(GIR == 1 ~ putts, TRUE ~ NA_real_),
               updown_conv = dplyr::case_when(GIR == 0 &
                                             par == gross ~ 1,
                                             TRUE ~ 0),
               updown_opps = dplyr::case_when(GIR == 0 &
                                             chips > 0 ~ 1,
                                             TRUE ~ 0)
  ) |>
  dplyr::rename(`Handicap Index` = handicap_index) |>
  dplyr::group_by(date, date_course, course_rating, `Handicap Index`) |>
  dplyr::summarize(
    FIRs = sum(FIR, na.rm = F),
    `Iron FIRs` = sum(`Iron FIRs`, na.rm = T),
    `Iron FIR%` = round(((sum(`Iron FIRs`, na.rm = T)/sum(`Iron FIR opps`, na.rm = T))*100), 1),
    `Driver FIRs` = sum(`Driver FIRs`, na.rm = T),
    `Driver FIR%` = round(((sum(`Driver FIRs`, na.rm = T)/sum(`Driver FIR opps`, na.rm = T))*100), 1),
    `FIR%` = round((sum(FIRs, na.rm = T)/sum(FIR_opps, na.rm = T))*100, 1),
    GIRs = sum(GIR, na.rm = F),
    `Par 3 GIRs` = sum(`Par 3 GIRs`, na.rm = T),
    `GIR%` = round((sum(GIRs, na.rm = T)/18)*100, 1),
    putts = sum(putts, na.rm = F),
    `Avg GIR putts` = round(mean(greenie_putts, na.rm = T), 2),
    chips = sum(chips, na.rm = F),
    `chips+putts` = sum(`chips+putts`, na.rm = F),
    `UpDown%` = round(((sum(updown_conv, na.rm = T)/sum(updown_opps, na.rm = T))*100), 1),
    pars = sum(is_gross_par, na.rm = F),
    birdies = sum(is_gross_birdie, na.rm = F),
  )

```

```

    bogies = sum(is_gross_bogey, na.rm = F),
    `doubles+` = sum(is_gross_bogey_worse, na.rm = F),
    penalties = sum(penalties, na.rm = T),
    `Gross Score` = mean(tot_gross, na.rm = F),
    `Net Score` = mean(tot_net, na.rm = F)) %>%
dplyr::mutate(dplyr::across(c(`Iron FIRs`, `Driver FIRs`, `Iron FIR%`, `Driver FIR%`, `FIR%`), ~dplyr::
  dplyr::across(c(`Par 3 GIRs`, `Avg GIR putts`, `UpDown%`, `GIR%`), ~dplyr::if_else(is.na(
    `chips+putts` = dplyr::case_when(is.na(chips) ~ NA_real_,
                                   TRUE ~ `chips+putts`)) %>%
dplyr::mutate(`UpAndDown%` = dplyr::case_when(grepl(date, pattern = '07-13|09-21') ~ NA, TRUE ~ `UpDown%`),
  `Iron FIR%` = dplyr::case_when(`Iron FIR` == NaN ~ 0.0, TRUE ~ `Iron FIR%`))

```

```

head(scores_sum |>
  dplyr::arrange(desc(date)))

```

```

## # A tibble: 6 x 26
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `Handicap Index`  FIRs `Iron FIRs` `Iron FIR%` `Driver
##   <date>    <chr>          <dbl>          <dbl> <int>      <dbl>      <dbl>      <dbl>
## 1 2026-04-05 "2026-04-05\nRando~      71.7           11      6         0         0
## 2 2026-03-29 "2026-03-29\nDell ~      70.3           10      5         2        66.7
## 3 2026-03-08 "2026-03-08\nRando~      71.7           10      2         0         0
## 4 2026-02-22 "2026-02-22\nDell ~      68            10.1       7         1        33.3
## 5 2026-02-08 "2026-02-08\nRando~      70            10.2       2         0         0
## 6 2026-01-25 "2026-01-25\nRando~      70            10.2       4         0        NaN
## # i 5 more variables: `doubles+` <int>, penalties <int>, `Gross Score` <dbl>, `Net Score` <dbl>, `Up

```

## View Metrics

Separate and view metrics

## Scoring Metrics Scores and Handicap

```

scoring_metrics <- scores_sum |>
  dplyr::select(`Handicap Index`, `Gross Score`, `Net Score`)
head(scoring_metrics |>
  dplyr::arrange(desc(date)))

```

```

## # A tibble: 6 x 6
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `Handicap Index`  `Gross Score` `Net Sc
##   <date>    <chr>          <dbl>          <dbl>      <dbl>      <dbl>
## 1 2026-04-05 "2026-04-05\nRandolph North\n11"      71.7           11         88
## 2 2026-03-29 "2026-03-29\nDell Urich\n10"      70.3           10         82
## 3 2026-03-08 "2026-03-08\nRandolph North\n10"      71.7           10         84
## 4 2026-02-22 "2026-02-22\nDell Urich\n10.1"      68            10.1       92
## 5 2026-02-08 "2026-02-08\nRandolph North\n10.2"      70            10.2       83
## 6 2026-01-25 "2026-01-25\nRandolph North\n10.2"      70            10.2       85

```

## Stroke Metrics Pars, birdies, bogies, etc.

```
stroke_metrics <- scores_sum |>
  dplyr::select(`doubles+`, bogies, pars, birdies)
head(stroke_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 7
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `doubles+` bogies  pars birdies
##   <date>    <chr>              <dbl>      <int>  <int> <int>  <int>
## 1 2026-04-05 "2026-04-05\nRandolph North\n11"      71.7         3     9     6     0
## 2 2026-03-29 "2026-03-29\nDell Urich\n10"      70.3         4     5     7     2
## 3 2026-03-08 "2026-03-08\nRandolph North\n10"      71.7         3     6     9     0
## 4 2026-02-22 "2026-02-22\nDell Urich\n10.1"        68          6     7     5     0
## 5 2026-02-08 "2026-02-08\nRandolph North\n10.2"      70          2     8     7     1
## 6 2026-01-25 "2026-01-25\nRandolph North\n10.2"      70          2     9     7     0
```

## Around-the-Green Metrics Chips, putts, etc.

```
atg_metrics <- scores_sum |>
  dplyr::select(chips, `chips+putts`, `UpAndDown%`, putts, `Avg GIR putts`)
head(atg_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 8
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating chips `chips+putts` `UpAndDown%` putts
##   <date>    <chr>              <dbl>  <dbl>      <dbl>      <dbl>  <int>
## 1 2026-04-05 "2026-04-05\nRandolph North\n11"      71.7    21         52         20    31
## 2 2026-03-29 "2026-03-29\nDell Urich\n10"      70.3    15         46         20    31
## 3 2026-03-08 "2026-03-08\nRandolph North\n10"      71.7    12         45         30    33
## 4 2026-02-22 "2026-02-22\nDell Urich\n10.1"        68     18         52        16.7   34
## 5 2026-02-08 "2026-02-08\nRandolph North\n10.2"      70     11         46        22.2   35
## 6 2026-01-25 "2026-01-25\nRandolph North\n10.2"      70     15         51        18.2   36
```

## Ball Striking Metrics Approach and tee accuracy

```
ball_striking_metrics <- scores_sum |>
  dplyr::select(GIRs, `GIR%`, `Par 3 GIRs`,
    FIRs, `FIR%`, `Iron FIRs`, `Iron FIR%`,
    `Driver FIRs`, `Driver FIR%`)
head(ball_striking_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 12
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating GIRs `GIR%` `Par 3 GIRs` FIRs `FIR%`
##   <date>    <chr>              <dbl>  <int>  <dbl>      <dbl>  <int>  <dbl>
## 1 2026-04-05 "2026-04-05\nRandolph North\n11"      71.7     3   16.7         1     6   42.9
## 2 2026-03-29 "2026-03-29\nDell Urich\n10"      70.3     7   38.9         1     5   38.5
```

```
## 3 2026-03-08 "2026-03-08\nRandolph North\n10" 71.7 7 38.9 2 2 14.3
## 4 2026-02-22 "2026-02-22\nDell Ulrich\n10.1" 68 3 16.7 1 7 53.8
## 5 2026-02-08 "2026-02-08\nRandolph North\n10.2" 70 7 38.9 3 2 14.3
## 6 2026-01-25 "2026-01-25\nRandolph North\n10.2" 70 6 33.3 2 4 28.6
```

**Club Metrics** Yardage and accuracy for each club

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date      club      n rd_min_yds_to_target rd_min_yds_traveled rd_min_yd_diff
##   <date>    <chr> <int>          <dbl>          <dbl>          <dbl>
## 1 2026-04-05 3W      2           250           222           -10
## 2 2026-04-05 4        4           115            82           -39
## 3 2026-04-05 5        1           205           210            -5
## 4 2026-04-05 6        4           190           165            2
## 5 2026-04-05 7        2           154           142            7
## 6 2026-04-05 8        1           165           167           -2
```

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date      club      n rd_max_yds_to_target rd_max_yds_traveled rd_max_yd_diff
##   <date>    <chr> <int>          <dbl>          <dbl>          <dbl>
## 1 2026-04-05 3W      2           250           260            28
## 2 2026-04-05 4        4           220           219            33
## 3 2026-04-05 5        1           205           210            -5
## 4 2026-04-05 6        4           190           188            25
## 5 2026-04-05 7        2           175           168            12
## 6 2026-04-05 8        1           165           167           -2
```

```
## # A tibble: 6 x 10
## # Groups:   date [1]
##   date      club `rd club strokes` miss_direction `rd club miss dir` rd_avg_yds_to_target rd_avg_yd_diff
##   <date>    <chr>          <int> <chr>          <int>          <dbl>
## 1 2026-04-05 3W      2 right          2           250
## 2 2026-04-05 4        4 right          3           151.
## 3 2026-04-05 4        4 short          1           151.
## 4 2026-04-05 5        1 long            1           205
## 5 2026-04-05 6        4 on_target       1           190
## 6 2026-04-05 6        4 right          1           190
```

## Plot Metric Summaries

### Scoring Metrics

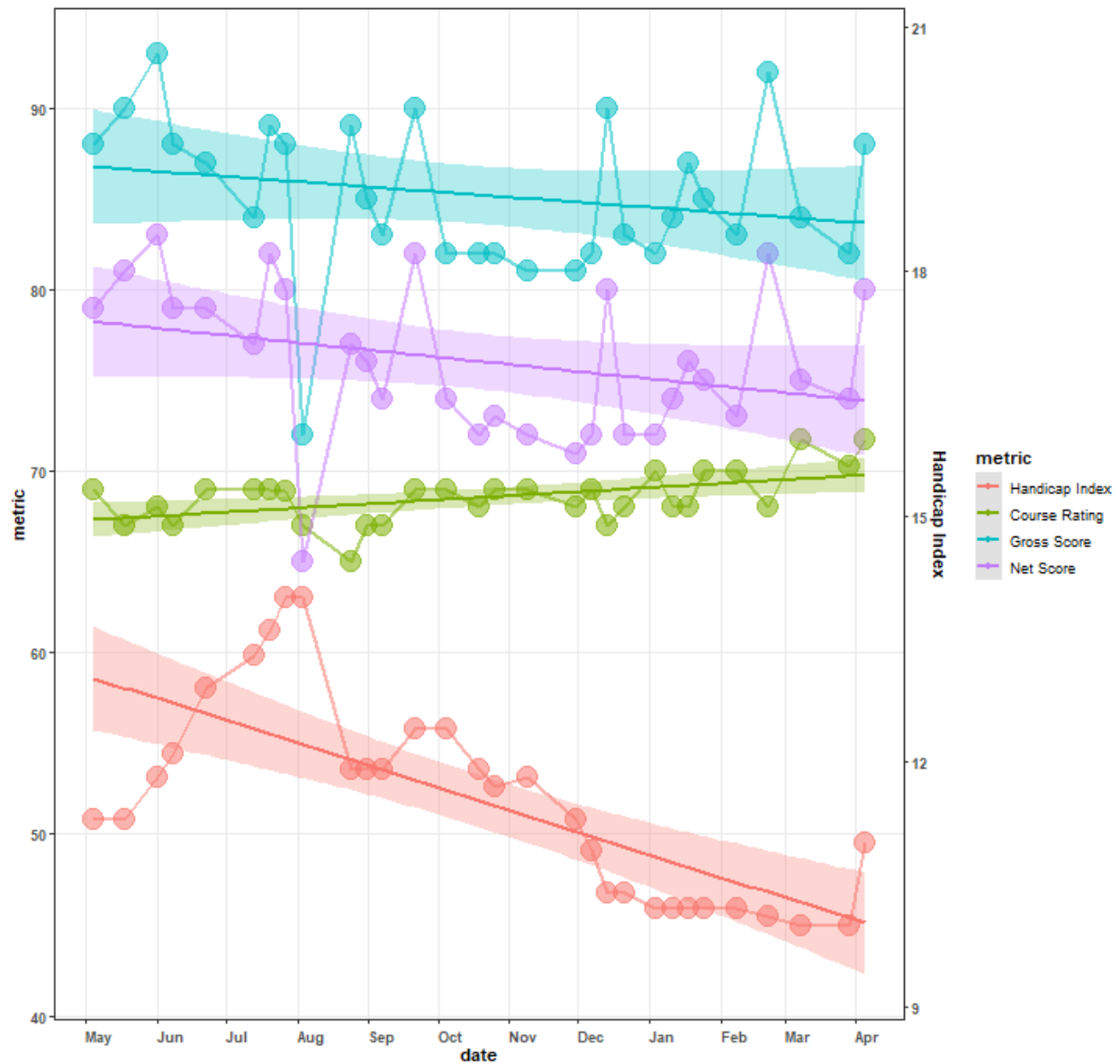
```
scoring_metrics |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
    `Handicap Index` = `Handicap Index`*4.5) |>
  dplyr::rename(`Course Rating` = course_rating) |>
  dplyr::group_by(date, date_course, `Handicap Index`) |>
  tidyr::pivot_longer(cols = c(`Course Rating`, `Net Score`), names_to = 'metric', values_to = 'value', v
  ggplot2::ggplot(aes(x = date_course, y = value)) +
```

```

ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1,
                        color = factor(metric,
                                      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
fill = factor(metric,
              levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1,
                      color = factor(metric,
                                    levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
fill = factor(metric,
              levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
ggplot2::geom_smooth(aes(x = date, y = value, group = metric,
                        color = factor(metric,
                                      levels = c('Handicap Index', 'Course Rating', 'Gross Score',
fill = factor(metric,
              levels = c('Handicap Index', 'Course Rating', 'Gross Score',
alpha = 0.3, method = 'lm')) +
ggplot2::scale_y_continuous(sec.axis = ggplot2::sec_axis(~./4.5, name = 'Handicap Index'))+
ggplot2::labs(title = paste0('Performance Over Time\n',
                             scores |>
                               dplyr::distinct(date) |>
                               dplyr::last() |>
                               unlist() %>% lubridate::as_date(.) |>
                               as.character(),
                             ' - ',
                             scores |>
                               dplyr::distinct(date) |>
                               dplyr::first() |>
                               unlist() %>% lubridate::as_date(.) |>
                               as.character()),
                  x = 'date',
                  y = 'metric',
                  color = 'metric'
                ) +
ggplot2::guides(alpha = 'none', size = 'none', fill = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 0, hjust = 0, vjust = 0),
               title = ggplot2::element_text(face = 'bold')
               ) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b')

```

Performance Over Time  
2025-05-04 - 2026-04-05



## Stroke Metrics

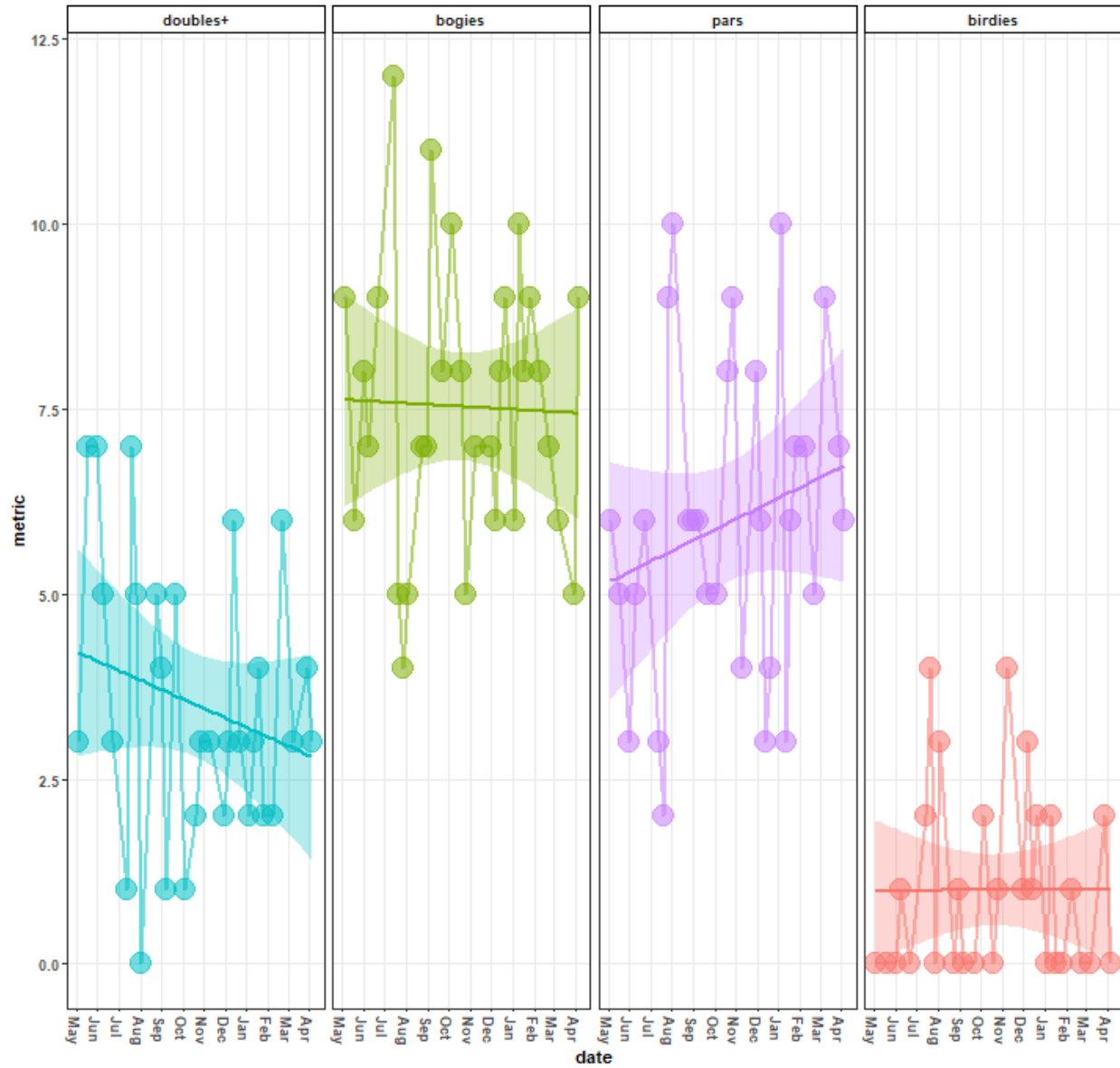
```
stroke_metrics |>
  tidyr::pivot_longer(cols = c(`doubles+`:birdies), names_to = 'metric', values_to = 'value', values_drop_na = TRUE) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Hole-by-Hole Scores Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('doubles+', 'bogies', 'pars', 'birdies'))), nrow = 1)

```

### Hole-by-Hole Scores Over Time 2025-05-04 - 2026-04-05



### Around the Green Metrics

```
atg_metrics |>
  tidyr::pivot_longer(cols = c(chips:`Avg GIR putts`), names_to = 'metric', values_to = 'value', values_from = 'score') +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Around the Green Metrics Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

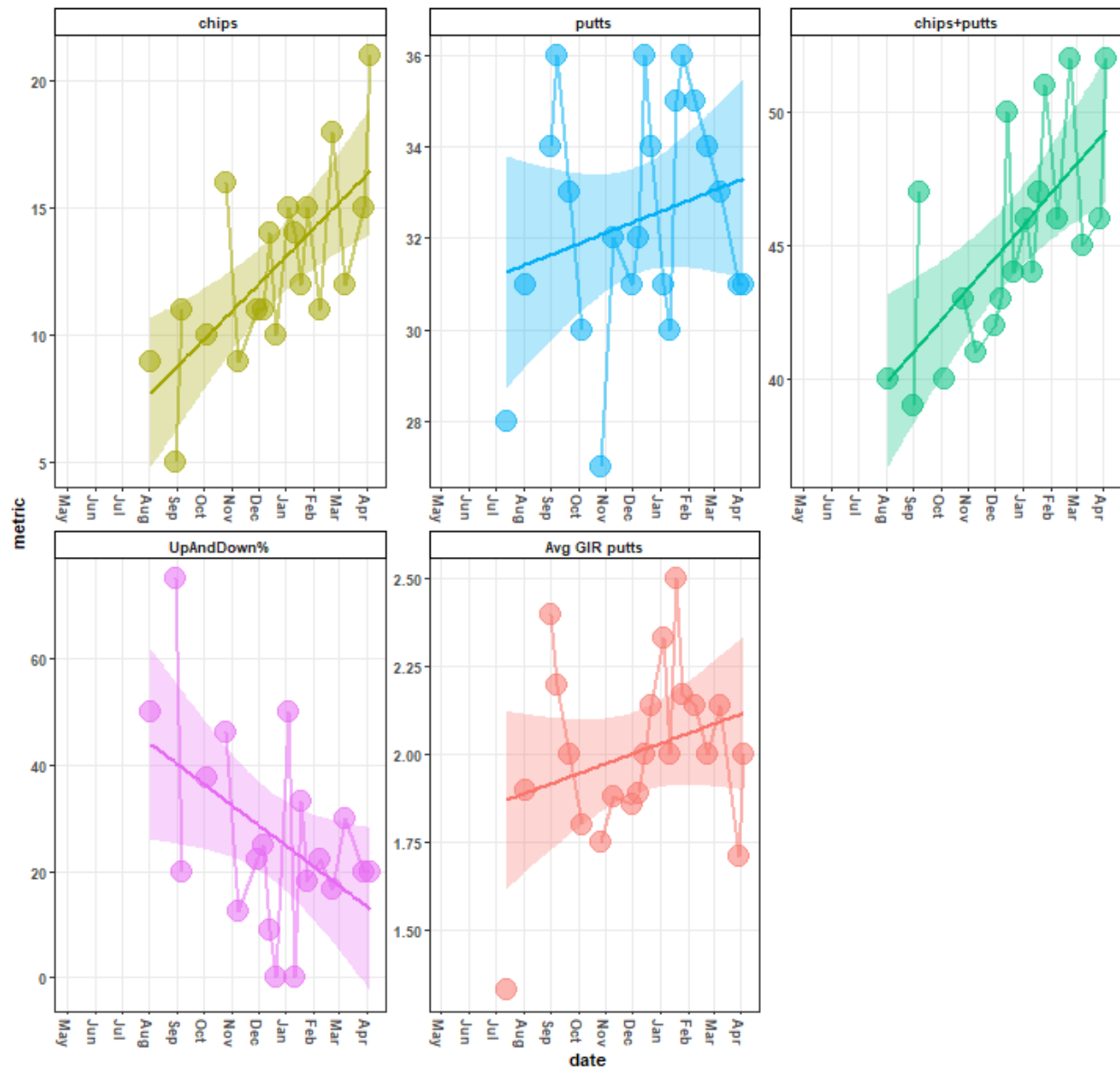


```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
    panel.grid.minor = ggplot2::element_blank(),
    axis.title = ggplot2::element_text(face = 'bold'),
    axis.text.y = ggplot2::element_text(face = 'bold'),
    axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
    title = ggplot2::element_text(face = 'bold'),
    strip.text.x.top = ggplot2::element_text(face = 'bold'),
    strip.background = ggplot2::element_rect(color = 'black',
        fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('chips', 'putts', 'chips+putts', 'UpAndDown%', 'Avg GI

```

## Around the Green Metrics Over Time 2025-05-04 - 2026-04-05



## Ball Striking Metrics

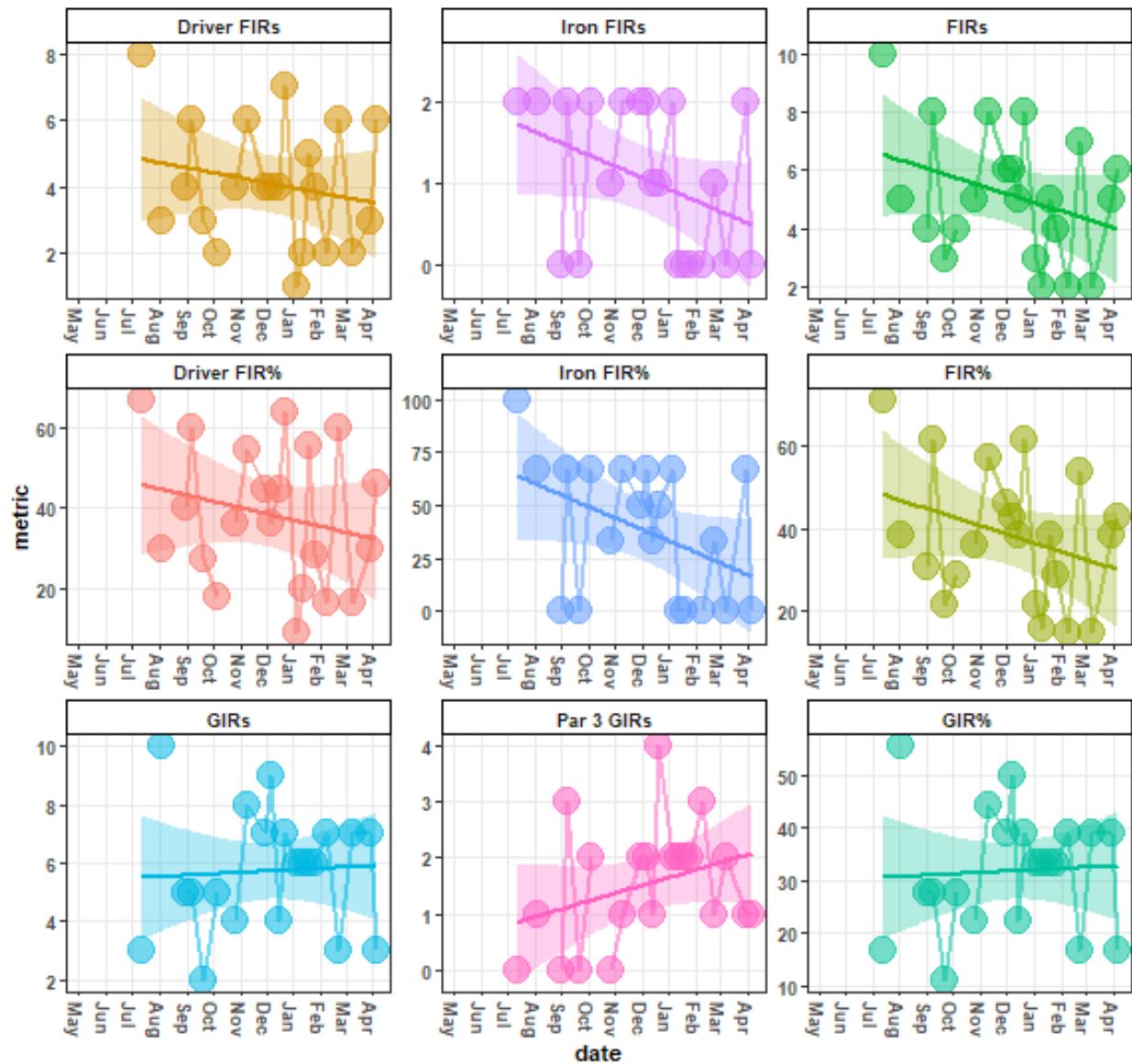
```
ball_striking_metrics |>
  tidyr::pivot_longer(cols = c(GIRs:`Driver FIR%`), names_to = 'metric', values_to = 'value', values_drop_na = T) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Ball Striking Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('Driver FIRs', 'Iron FIRs', 'FIRs',
                                              'Driver FIR%', 'Iron FIR%', 'FIR%',
                                              'GIRs', 'Par 3 GIRs', 'GIR%')), scales = 'free')

```

## Ball Striking Over Time 2025-05-04 - 2026-04-05



## Stroke Quality Metrics

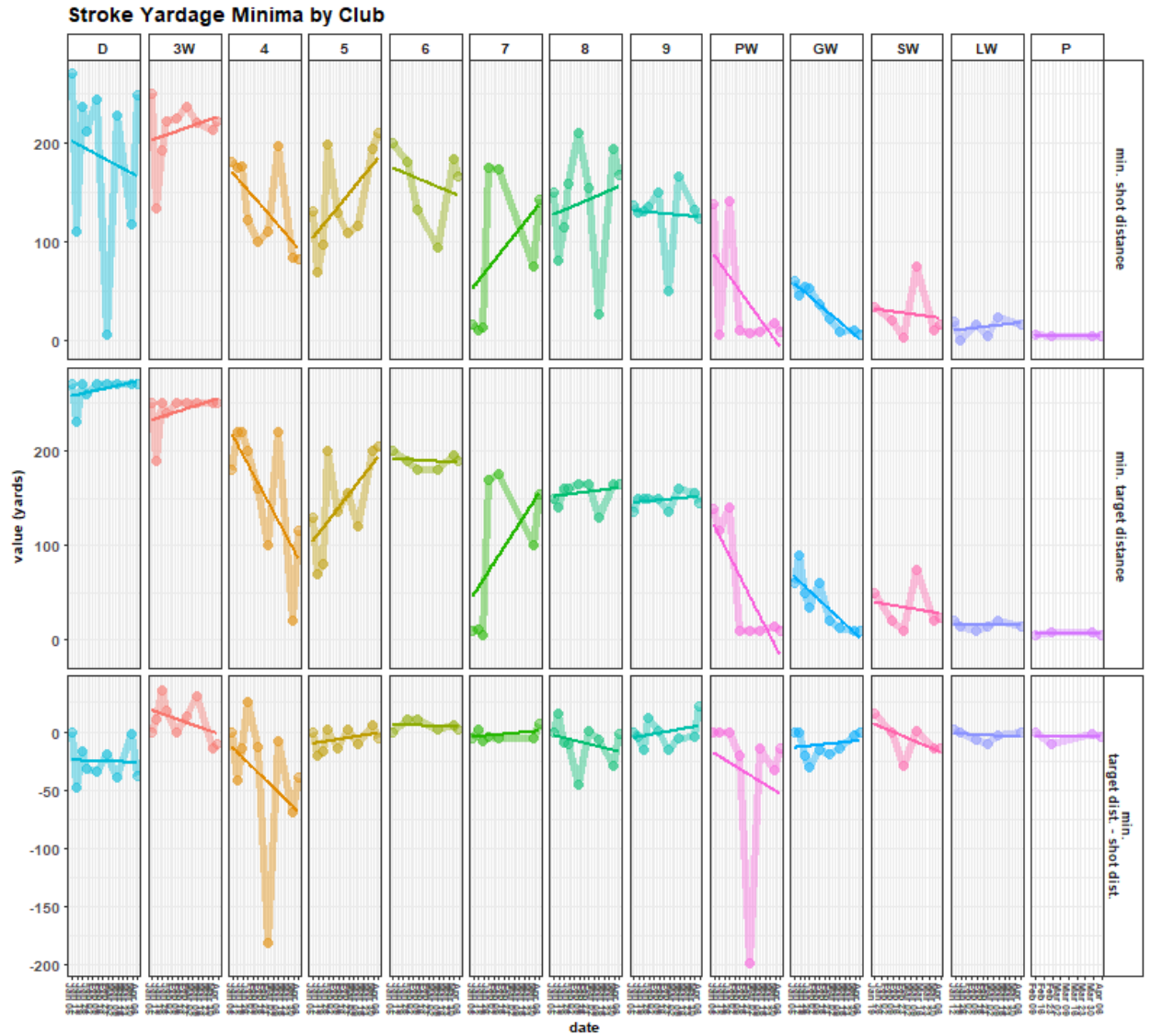
```
stroke_quality_min |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(
    grepl(metric, pattern = 'target') ~ 'min. target distance',
    grepl(metric, pattern = 'traveled') ~ 'min. shot distance',
    grepl(metric, pattern = 'diff') ~ 'min. \ntarget dist. - shot d

  dplyr::mutate(date = lubridate::as_date(date),
    club = as.character(club),
```

```

n = as.integer(n)) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Stroke Yardage Minima by Club', x = 'date', y = 'value (yards)', fill = 'club')
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(rows = vars(metric), cols = vars(factor(club,
                                                           levels =
                                                             c('D', '3W', '4',
                                                             '5', '6', '7',
                                                             '8', '9', 'PW',
                                                             'GW', 'SW', 'LW',
                                                             'P'))),
                    scales = 'free')

```



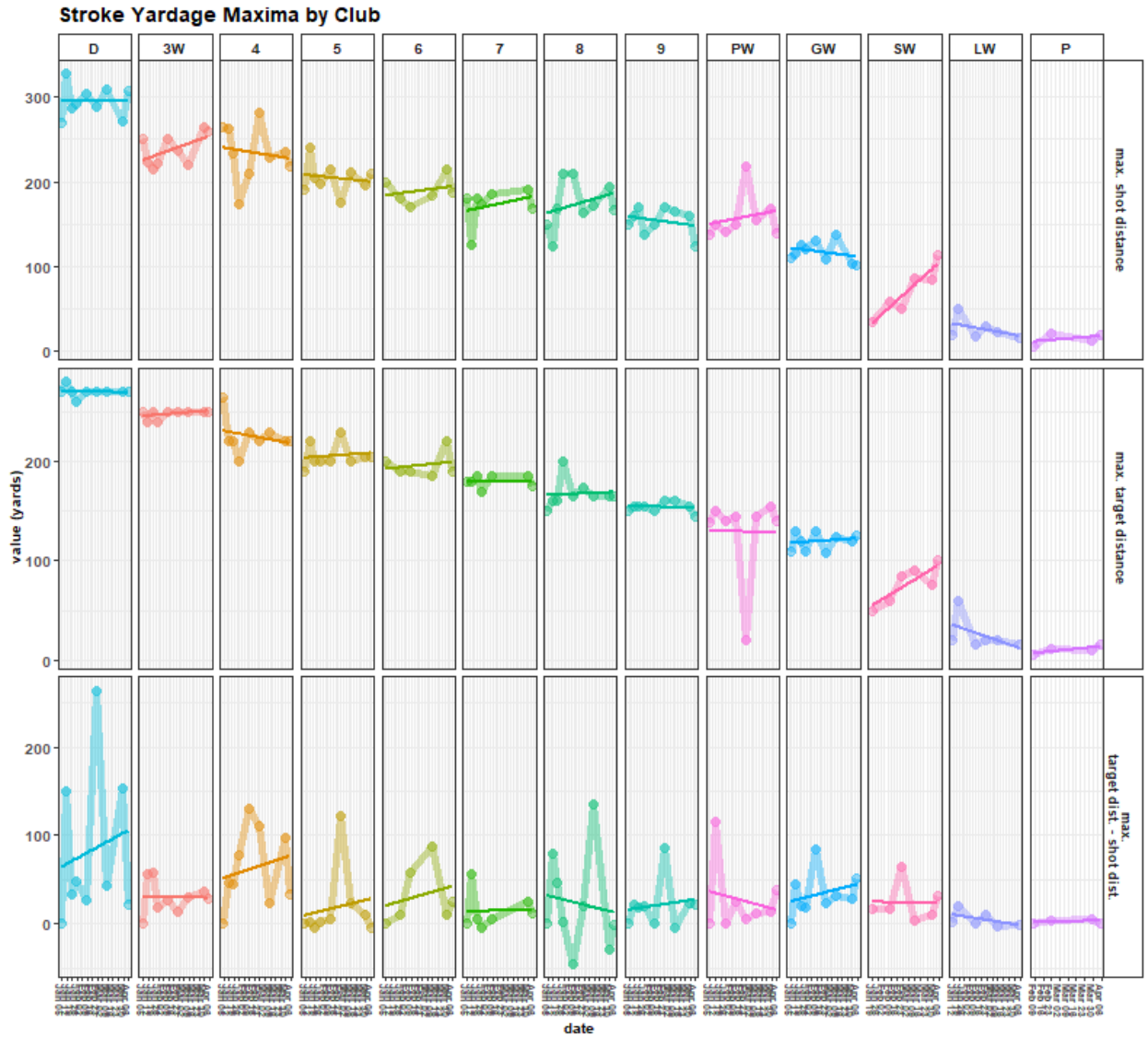
Minima

```
stroke_quality_max |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'max. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'max. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'max.\ntarget dist. - shot dist.'))
  dplyr::mutate(date = lubridate::as_date(date),
                 club = as.character(club),
                 n = as.integer(n)) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Stroke Yardage Maxima by Club', x = 'date', y = 'value (yards)', fill = 'club')
```

```

ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(rows = vars(metric), cols = vars(factor(club,
                                                           levels =
                                                             c('D', '3W', '4',
                                                               '5', '6', '7',
                                                               '8', '9', 'PW',
                                                               'GW', 'SW', 'LW',
                                                               'P'))),
                   scales = 'free')

```



Maxima

```
stroke_quality_avg |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(dplyr::everything(), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains("rd")), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'avg. target distance',
    grepl(metric, pattern = 'traveled') ~ 'avg. shot distance',
    grepl(metric, pattern = 'diff') ~ 'avg.\ntarget dist. - shot dist.',
    # grepl(metric, pattern = 'accuracy') ~ 'avg\naccuracy',
    grepl(metric, pattern = 'strokes') ~ 'n strokes',
    grepl(metric, pattern = 'dir') ~ 'strokes w/club\nin round\nby',
    grepl(metric, pattern = 'acc') ~ '% accuracy')) |>

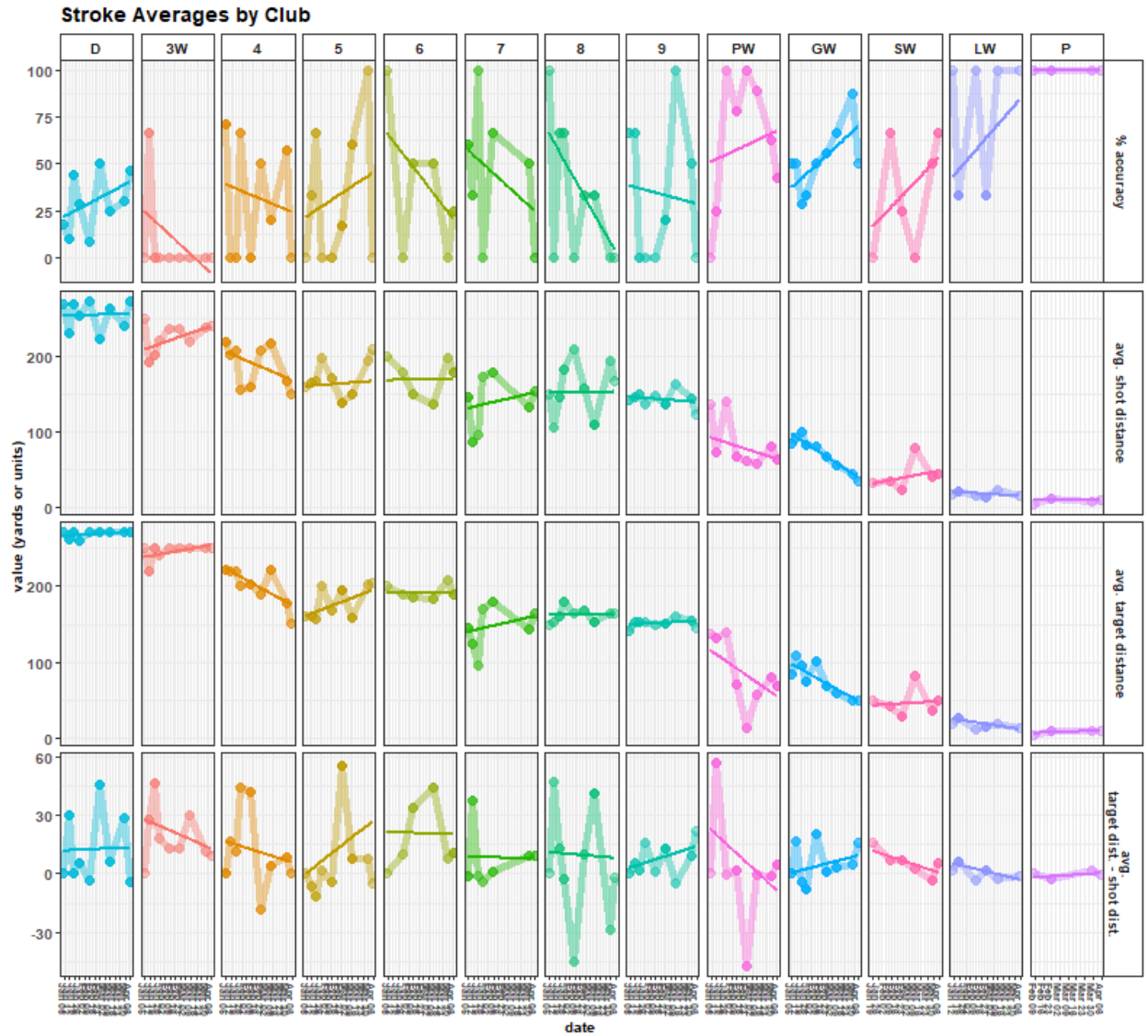
  dplyr::filter(!grepl(metric, pattern = 'strokes')) |>
  dplyr::mutate(date = lubridate::as_date(date),
    club = as.character(club),
    vals = as.numeric(vals))
```



```

) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Stroke Averages by Club', x = 'date', y = 'value (yards or units)', fill = 'club') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(rows = vars(metric),
                   cols = vars(factor(club,
                                     levels = c('D', '3W', '4', '5', '6',
                                                '7', '8', '9', 'PW', 'GW',
                                                'SW', 'LW', 'P'))),
                   scales = 'free')

```



Average

Main Metrics

```
scores_sum |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
  tidyr::pivot_longer(cols = c(`Handicap Index`, `Gross Score`, `Net Score`,
    `doubles+`, bogies, pars, birdies, chips, putts,
    `UpDown%`, `Avg GIR putts`, `GIR%`, `Par 3 GIRs`,
    `Iron FIR%`, `Driver FIR%`), names_to = 'metric', values_to = 'value', v
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Performance Over Time\n',
    scores |>
      dplyr::distinct(date) |>
      dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
  panel.grid.minor = ggplot2::element_blank(),
  axis.title = ggplot2::element_text(face = 'bold'),
  axis.text.y = ggplot2::element_text(face = 'bold'),
  axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
  title = ggplot2::element_text(face = 'bold'),
  strip.text.x.top = ggplot2::element_text(face = 'bold'),
  strip.background = ggplot2::element_rect(color = 'black',
    fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric,
  levels = c('Handicap Index', 'Gross Score', 'Net Score',
    'doubles+', 'bogies', 'pars', 'birdies',
    'chips', 'putts', 'UpDown%', 'Avg GIR putts',
    'GIR%', 'Par 3 GIRs', 'Iron FIR%', 'Driver FIR%')),
  scales = 'free',
  ncol = 4)

```

# Performance Over Time 2025-05-04 - 2026-04-05

